

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE**III Semester – MCA - C2 Internal Assessment, Febraury - 2026****Subject: Machine Learning****Time: 75 Minutes****Max. Marks: 30****PART A****Answer the following** **$2 \times 10 = 20$**

- 1 a. Explain Kmeans algorithm with an example.
- b. Discuss the major clustering approaches.

CO3 06

CO3 04

OR

- 2 a. Explain PCA in detail. Calculate the principal component for the given data

CO3 10

Size (in canal) X_1	No. of Rooms X_2
5	10
3	7
10	15
2	4

- 3 Using Bayesian Network calculate the probability that a player plays tennis or not upon the condition $x = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$ for the given dataset

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

a.

CO4 10

OR

- 4 a. Explain Markov Model with an example.

CO4 07

- b. Discuss the types of Naïve Bayes Classifier.

CO4 03

PART B

Answer the following

$2 \times 5 = 10$

- 5 Write a note on Similarity Measures in clustering.

CO3 5

OR

- 6 Explain the Criteria functions for Clustering.

CO3 5

- 7 Explain inference in Probabilistic Graphical Models.

CO4 5

OR

- 8 Discuss the learning generalization in ML.

CO4 5

St. Philomena's College (Autonomous), Mysore
III Semester MCA C2 Internal Assessment Test, February-2026

Title: Python Programming

Time: 75 minutes

Max Marks: 30

PART-A

Answer the following

2×10=20

1. a. i. Define a class in Python. Explain the role of constructor and destructor with suitable example. CO:CO3
ii. What is the purpose of self and del keywords in Python? Illustrate with example. (06 + 04)

OR

- b. i. Explain data attributes and class attributes. Write a Python program demonstrating both. CO:CO3
ii. Explain inheritance in Python with suitable example. (06 + 04)
2. a. i. Explain file object attributes in Python. Write a Python program to read and display the contents of a file. CO:CO4
ii. Explain rename and delete file operations in Python with example. (06 + 04)

OR

- b. i. Define Regular Expressions. Explain any four regular expression metacharacters with examples. CO:CO4
ii. Write a Python program using regular expressions to extract all numbers from a given string. (06 + 04)

PART-B

Answer the following

2×5=10

3. a. Explain exception handling in Python. Write a Python program demonstrating user-defined exception. CO:CO3

OR

- b. Explain static members in Python with suitable example. CO:CO3
4. a. Explain Tkinter widgets. Write a Python program to create a simple GUI window with label and button. CO:CO4

OR

- b. Explain unit testing in Python. Write a simple test case using unittest. CO:CO4

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester – MCA – C2 Internal Assessment Test: February - 2026

Subject: Mobile Application Development

Time: 75 Minutes

Max. Marks: 30

PART A

Answer the following

2 × 10 = 20

- 1
- a. Explain navigation in Flutter using the Navigator class. Describe push() and pop() methods with example. CO3 6
 - b. Explain named routes in Flutter and discuss their advantages over direct navigation. CO3 4

OR

- 2
- a. Explain responsive design in Flutter and discuss the importance of MediaQuery with suitable example CO3 6
 - b. Explain how Flutter packages are used. Describe the steps to install and use a package from pub.dev. CO3 4
- 3
- a. Explain RESTful APIs and their role in mobile application development. CO4 6
 - b. Explain how HTTP requests are made in Flutter using the http package with example. CO4 4

OR

- 4
- a. Explain asynchronous programming in Flutter. Describe Future, async, and await with example. CO4 6
 - b. Explain JSON parsing in Flutter and discuss serialization and deserialization. CO4 4

PART B

Answer the following

2 × 5 = 10

- 5
- Explain any five commonly used Flutter packages such as image_picker, url_launcher, flutter_svg, http, intl, loading_animation_widget, google_maps_flutter, or geolocator. CO3 5

OR

- 6
- Explain how Google Maps and location services can be integrated in Flutter using google_maps_flutter and geolocator packages. CO3 5
- 7
- Explain state management in Flutter and discuss its purpose and benefits. CO4 5

OR

- 8
- Explain any two state management approaches in Flutter such as Provider, Riverpod, or BLoC. CO4 5

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE
III Semester – MCA – C2 Internal Assessment Test: February - 2026

SUBJECT: ETHICAL HACKING

Time: 75 Minutes

Max. Marks: 30

PART A

Answer the following

2 × 10 = 20

- | | | | |
|---|---|-----|---|
| 1 | a. Explain Denial of Service (DoS) attacks and their impact on system availability. | CO3 | 6 |
| | b. List and explain countermeasures against DoS attacks. | CO3 | 4 |

OR

- | | | | |
|---|---|-----|---|
| 2 | a. Explain web server and application vulnerabilities with suitable examples. | CO3 | 6 |
| | b. Describe SQL Injection attacks and their consequences. | CO3 | 4 |
| 3 | a. Explain buffer overflow attacks with a neat diagram. | CO4 | 6 |
| | b. What is vulnerability analysis? Explain its importance in penetration testing. | CO4 | 4 |

OR

- | | | | |
|---|--|-----|---|
| 4 | a. Explain client-side browser exploits and their impact on end users. | CO4 | 6 |
| | b. Describe reverse engineering and its role in vulnerability discovery. | CO4 | 4 |

PART B

Answer the following

2 × 5 = 10

- | | | | |
|---|--|-----|---|
| 5 | Explain the Windows Access Control Model and how it can be exploited for local privilege escalation. | CO4 | 5 |
|---|--|-----|---|

OR

- | | | | |
|---|--|-----|---|
| 6 | Explain local elevation of privilege attacks with suitable examples. | CO4 | 5 |
| 7 | Explain vulnerabilities in mobile applications. | CO4 | 5 |

OR

- | | | | |
|---|---|-----|---|
| 8 | Describe techniques used to exploit vulnerabilities in mobile applications. | CO4 | 5 |
|---|---|-----|---|

St. Philomena's College (Autonomous), Mysore
III Semester MCA C2 Internal Assessment Test, February-2026

Title: Software Engineering

Time: 75 minutes

Max Marks: 30

PART-A

Answer the following

2×10=20

1. a. Discuss factors affecting software pricing.

CO:CO3

OR

- b. i. Discuss the various inspection checks in program inspection.
ii. What are the two distinct goals of software testing?

CO:CO3

(06 + 04)

2. a. Explain the design guidelines for developing secure systems.

CO:CO4

OR

- b. Briefly explain the three logical stages in service engineering process.

CO:CO4

PART-B

Answer the questions:

2×5=10

3. a. Explain interface testing with an example.

CO:CO3

OR

- b. Write any five product and process standards.

CO:CO3

4. a. Write the terminologies used in aspect-oriented software development.

CO:CO4

OR

- b. Briefly explain the stages in survivability analysis.

CO:CO4

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester – MCA – C2 Internal Assessment: February - 2026

Subject : DIGITAL IMAGE PROCESSING

Time: 75 Minutes

Max. Marks: 30

PART A

Answer the following:

2 × 10 = 20

- 1 a. Discuss the role of Thresholding in image segmentation. CO3 10

OR

- 2 a. Discuss the working of Region growing and Region Split & Merge approaches in image segmentation. CO3 10

- 3 a. Explain erosion & dilation in morphological image processing with suitable diagrams & examples. CO4 10

OR

- 4 a. What is robot vision? Discuss various real-world applications of robot vision. CO4 10

PART B

Answer the following:

2 × 5 = 10

- 5 Explain how Canny edge detection algorithm is used to detect edges in image segmentation. CO3 5

OR

- 6 Write a note on the use of motion in image segmentation. CO3 5

- 7 Differentiate between opening and closing in morphological image processing. CO4 5

OR

- 8 Explain hole filling in morphological image processing with suitable example. CO4 5
